

Maheen Naeem

☎ +92-3090054878 ✉ maheennaem94@gmail.com 🔗 linkedin.com/in/maheen-naeem
🐙 github.com/MaheenNaeem29 🌐 portfolio-beryl-tau-31.vercel.app

PROFESSIONAL SUMMARY

Artificial Intelligence practitioner specializing in Multimodal Deep Learning, Computer Vision, and AI-driven Security Automation. Experienced in developing scalable end-to-end machine learning systems, from data preprocessing and model training to deployment. Proficient in deploying production-ready applications using FastAPI and Docker.

TECHNICAL SKILLS

Languages	Python, C++, SQL, JavaScript, React, Bash
AI/ML	CNNs, Transformers, YOLOv8, NLP (Hugging Face), Multimodal Learning, SHAP, LIME
Frameworks	PyTorch, TensorFlow, Scikit-Learn, OpenCV, Pandas, NumPy
Tools	Docker, FastAPI, Redis, Celery, AWS (EC2/S3), Git, Linux
Security	Nmap, Metasploit, OpenVAS

SELECTED PROJECTS

ASHEN – AI-Driven Automated Penetration Testing *Final Year Project*

- Engineered an automated security framework using LLMs to analyze vulnerability scans, reducing manual reporting time by 65%.
- Architected a distributed backend using FastAPI and Redis, enabling concurrent host exploitation and real-time monitoring.
- Implemented automated remediation generation, achieving 95% accuracy in providing actionable security patches.

Multimodal Cancer Prognosis System *Deep Learning Specialist*

- Developed a Late-Fusion Neural Network integrating Histopathology, Genomic, and Clinical data, achieving a 0.89 AUC-ROC.
- Leveraged Explainable AI (XAI) to visualize feature importance, ensuring model decisions are clinically interpretable.
- Optimized data preprocessing pipelines in PyTorch to handle heterogeneous data sources across multiple cancer types.

Smart City Urban Surveillance *Computer Vision Project*

- Fine-tuned YOLOv8 for real-time pedestrian detection, attaining 83% Precision on custom-annotated urban datasets.
- Deployed models using TensorRT for optimized inference, maintaining 30+ FPS on edge-level hardware.

CERTIFICATIONS

- Machine Learning Specialization – Stanford Online (DeepLearning.AI)
- AWS Certified Cloud Practitioner – *In Progress*

EDUCATION

Bachelor of Science in Artificial Intelligence *2022 – 2026 (Expected)*

FAST-NUCES, Chiniot-Faisalabad

Relevant Coursework: Deep Learning, Computer Vision, Probabilistic Graphical Models, Big Data Analytics

SOFT SKILLS

TimeManagement	Balanced full-time academic coursework load
Communication	Presented complex technical research findings and conducted lab demonstrations
Problem Solving	Debugged complex deep learning pipelines and optimized model performance

LANGUAGES

English: Fluent (Professional Proficiency)